Geometric Progression (गुणोत्तर श्रेणी):

- **General form:** $a,\ ar,\ ar^2, ar^3, \ldots\ldots\ldots$
- **(n^{th} term)** $T_n = a \cdot r^{\,n-1}$
- **(Sum of n terms)** $S_n = \frac{a\ r^n - 1}{r-1} = \frac{a\ 1-r^n}{1-r}$
- **(Sum of infinite terms)** $S_\infty = \frac{a}{1-r}, \quad -1 < r < 1$

$20, -10, 5, -\frac{5}{2} \ldots\ldots\ldots$

$a = 20$

$r = -\frac{1}{2}$

→ Common ratio

सार्व अनुपात

$a=3 \quad r=2$

$3, 6, 12, 24 \ldots$

$T_8 = ar^7 = 3 \times 2^7$

$S_{10} = \frac{a(r^n - 1)}{r-1} = \frac{3(2^{10} - 1)}{2-1}$

Decreasing GP



$12, 6, 3, \frac{3}{2}, \ldots$ $a=12$ $r=\frac{1}{2}$

$S_n = \frac{a(1-r^n)}{1-r}$

$S_\infty = \frac{a(1-r^\infty)}{1-r}$

$S_\infty = \frac{a}{1-r}$

$\left(\frac{1}{2}\right)^1 = \frac{1}{2}$

$\left(\frac{1}{2}\right)^2 = \frac{1}{4}$

$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$

$\left(\frac{1}{2}\right)^{1000} =$

$\left(\frac{1}{2}\right)^{\infty} = 0$

$3, 6, 12, 24, \ldots\ldots$ $a=3, r=2$

$3, -6, 12, -24, 48, \ldots\ldots$ $a=3, r=-2$

$\rightarrow S_\infty = \infty$

$S_\infty = \pm\infty$

$S_\infty = -\infty$

$-3, -6, -12, -24, \ldots\ldots$ $a=-3$ $r=2$

Increasing GP

Geometric mean (:

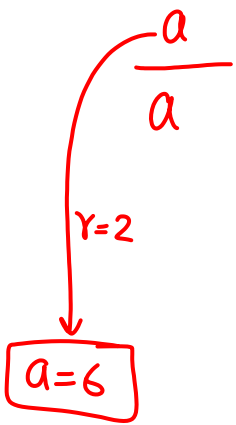
- **Geometric mean of**

$$a_1, a_2, a_3, \ldots\ldots, a_n = \sqrt[n]{a_1 \times a_2 \times a_3 \times \ldots\ldots\ldots \times a_n}$$

- **Geometric mean of a, b is $\sqrt{ab}$.**
- **If a, b, c are in GP then $b^2 = ac$**

21. In a G.P., the 3rd term is 24 and the 6th term is 192. Find the 10th term.

a) 512 b) 1536
c) 3072 d) 6144



22. The 5th , 8th and 11th terms of a G.P. are p, q and s, respectively. Find q^2.

किसी गुणोत्तर श्रेणी का 5वीं, 8वाँ तथा 11वाँ पद क्रमश: p, q तथा s है तो दिखाइए कि $q^2 = ps$

a) $p + s$ b) p/s

c) ps d) s/p

AP

$7, 10, 13, 16, 19, \ldots$ (3, 3)

1st 3rd 5th

7 — 6 — 13 — 6 — 19

AP

$10^{th}, 15^{th}, 20^{th} \rightarrow$ AP

GP

$3, 6, 12, 24, 48, 96, \ldots$

1st 3rd 5th

3 12 48

$12^{th}, 15^{th}, 18^{th} \rightarrow$ GP ✓

23. The 4th term of a G. P. is square of its second term, and the first term is -3. Determine its. 7th term.
किसी गुणोत्तर श्रेणी का चौथा पद उसके दूसरे पद का वर्ग है तथा प्रथम पद -3 है तो 7वाँ पद ज्ञात कीजिए।

a) -2187 b) -343
c) -729 d) 343

24. **The sum of first three terms of a G.P. is 16 and the sum of the next three terms is 128. Determine the first term.**

a) 2

b) 16

c) $\frac{16}{7}$

d) $\frac{7}{16}$

25. **Find four numbers forming a geometric progression in which the third term is greater than the first term by 9, and the second term is greater than the 4th by 18.**

26. **The second term of a geometric progression is 12 more than the first term and the third term is 28 more than the second. What is the difference between the third and fourth terms of that progression?**

a) 44 b) 65.33

c) 60 d) 80

27. The sum of first three terms of a G.P. is $\frac{39}{10}$ and their product is 1. Find the common ratio.

एक गुणोत्तर श्रेणी के तीन पदों का योगफल $\frac{39}{10}$ गुणोत्तर हैं तथा उनका गुणनफल 1 है| सार्व अनुपात ज्ञात कीजिए।

a) $\frac{5}{2}$ b) $\frac{2}{5}$

c) $\frac{5}{2}$ or $\frac{2}{5}$ d) 1

AP → $a-d,\ a,\ a+d$

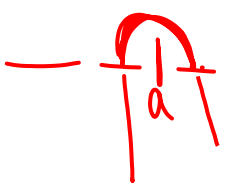
$a-2d,\ a-d,\ a,\ a+d,\ a+2d$

$a-3d,\ a-d,\ a+d,\ a+3d$

GP → $\frac{a}{r},\ a,\ ar$

$\frac{a}{r^2},\ \frac{a}{r},\ a,\ ar,\ ar^2$

$\frac{a}{r^3},\ \frac{a}{r},\ ar,\ ar^3$



Que.27

$$\frac{a}{r} \times a \times ar = 1$$

$a^3 = 1$

$a = 1$

$a = 1$

$$\frac{1}{r} + 1 + r = \frac{39}{10}$$

$$\frac{1+r^2}{r} = \frac{29}{10}$$

28. If a, b, c are in G.P. and $a^{\frac{1}{x}} = b^{\frac{1}{y}} = c^{\frac{1}{z}}$, then x, y, z are in

c^{-}

a) AP
b) GP
c) HP
d) None

29. The sum of some terms of G.P. is 315 whose first term and the common ratio are 5 and 2, respectively. Find the last term.

गुणोत्तर श्रेणी के कुछ पदों का योग 315 है, उसका प्रथम पद तथा सार्व अनुपात क्रमश: 5 तथा 2 हैं। अंतिम पद ज्ञात कीजिए।

a) 80 b) 160

c) 180 d) 320

30. If the arithmetic and geometric mean of two numbers are 12.5 and 12 respectively, find the difference of the two numbers?

a) 0.5 b) 5

c) 6 d) 7

31. If A.M. and G.M. of roots of a quadratic equation are 8 and 5, respectively, then obtain the quadratic equation.

32. If p, q, r, s are in geometric progression, then $(p^2 + q^2 + r^2)(q^2 + r^2 + s^2) =$

यदि p, q, r, s गुणोत्तर श्रेणी में हैं, तो $(p^2 + q^2 + r^2)(q^2 + r^2 + s^2) =$

a) $pqrs$

b) $p + q + r + s$

c) $(pq + qr + rs)^2$

d) $p^2 + q^2 + r^2 + s^2$

33. If p, q, r, s are in geometric progression, then $(p^3 + q^3)$, $(q^3 + r^3)$, $(r^3 + s^3)$ are in:
यदि p, q, r, s गुणोत्तर श्रेणी में हैं, तो $p^3 + q^3$, $(q^3 + r^3)$, $(r^3 + s^3)$ निम्न में हैं:

a) A.P
b) G.P
c) H.P
d) None

$$1+2+3+4+\ldots+n=\sum n=\frac{n(n+1)}{2}$$

$$1^2+2^2+3^2+\ldots+n^2=\sum n^2=\frac{n(n+1)(2n+1)}{6}$$

$$1^3+2^3+3^3+\ldots+n^3=\sum n^3=\left[\frac{n(n+1)}{2}\right]^2$$

$$1\times2+2\times3+3\times4+\ldots+n(n+1)=\sum n(n+1)=\frac{n(n+1)(n+2)}{3}$$

$$1\times2\times3+2\times3\times4+3\times4\times5+\ldots+n(n+1)(n+2)$$

$$=\sum n(n+1)(n+2)=\frac{n(n+1)(n+2)(n+3)}{4}$$

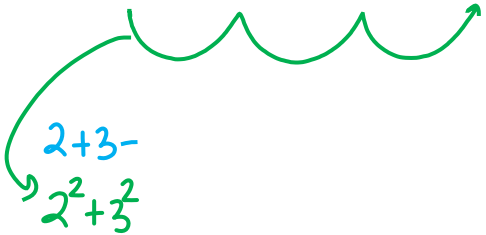
Eg: $1\times2+2\times3+3\times4+\ldots+\underset{\downarrow}{10}\times11=\frac{10\times11\times12}{3}=440$





34. **Find the sum of the first 40 terms of $2+4+3+9+4+16+\ldots\ldots$.**
$2+4+3+9+4+16+\cdots$ के पहले 40 पदों का योग ज्ञात कीजिए

a) 2048 b) 3310
c) 3540 d) 3542



$$\left[1+2+3+\ldots+21\right] - \left[1^2+2^2+\ldots+21^2\right] - 2 \quad A$$

$$= \frac{7 \quad 11}{\cancel{21} \times \cancel{22} \times} \\ \cancel{2} \quad \cancel{3}$$

$$=$$

):

- **General form:** $\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \frac{1}{a+3d}, \ldots\ldots\ldots$
- **(n^{th} term)** $T_n = \frac{1}{a+\ n-1\ d}$

Harmonic mean (:

- **Harmonic mean of**

$$a_1, a_2, a_3, \dots\dots, a_n = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \cdots \dots + \frac{1}{a_n}}$$

- **Harmonic mean of a, b is** $\frac{2}{\frac{1}{a}+\frac{1}{b}} = \frac{2ab}{a+b}$

M

$$\frac{2ab}{a+b}$$

35. There are two numbers. If the ratio of their arithmetic mean to harmonic mean is $25 : 9$, what is the ratio of their arithmetic mean to geometric mean?

a) $3 : 5$ b) $5 : 3$

c) $9 : 25$ d) $25 : 9$